Urban Development Forecast in Atlanta Metropolitan Statistical Area

—Richard Barad, Jonathan Manurung

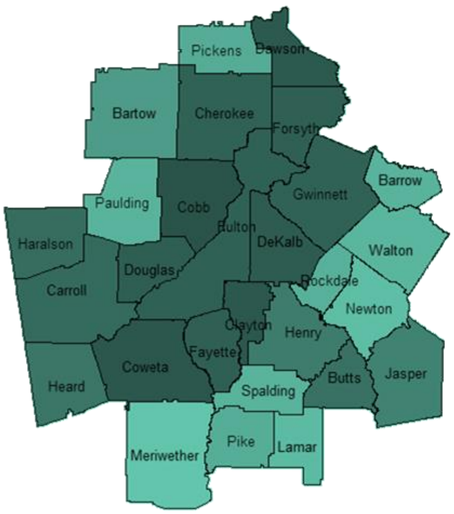

Figure 1. Atlanta MSA

Atlanta-Sandy-Springs-Roswell Metropolitan Statistical Area (MSA) is one of the fastest growing metropolitan areas in the United States. By 2031, the population is forecasted to reach 6,662,000 people. This high population growth has resulted in an increase in development, which lead to higher likelihood of urban sprawl and may have detrimental impacts on the environment. This analysis try to develop a model based on where development took place between 2011 and 2021 and forecast where development will likely take place between 2021 and 2031.

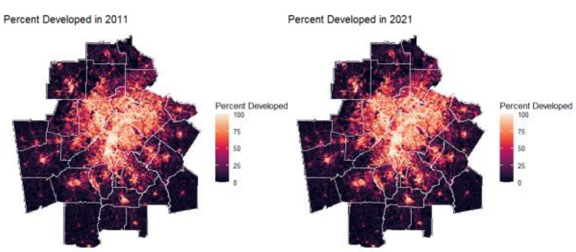

Figure 2. Percent Developed Grid Square

Land Cover Data used in this analysis comes from National Land Cover Database (NLCD) for 2011 and 2021. The land cover rasters include 15 land cover categories, which then reclassified into five general categories, Developed, Forest, Farm, Wetlands, Other Undeveloped. Figure 2 explains the percent of each grid square that is developed in 2011 and 2021 with brighter color grid squares represent more developed and darker color grid squares represent not developed. In this analysis, a grid square is classified as developed if it is more than 25% developed.

Grid square is associated with a county by calculating the overlap with county boundaries. Grid squares that are water and are already developed are removed from before training the model

EXPLORATORY ANALYSIS

The next part of the analysis is exploring the data. Figure 3 shows the number of grid squares in the dataset for each categories.

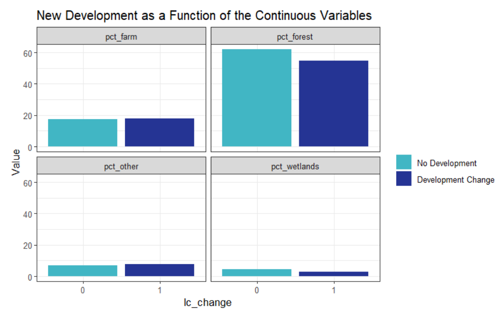

Figure 3. Development Based on Land Cover Categories

MODELING

Predictors are added to the grid square in order to train the model. Land Cover, Distance to Nearest Developed Cell, Population Data, and Distance to Highway are the predictors used in this model training.

Five Different Models are trained distinguishing between each predictor and the spatial lag. Figure 4 shows the model comparison

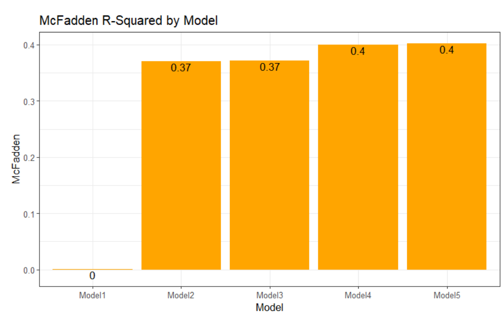

Figure 4. Model Comparison

A data frame is created including columns for the observed development change, 'lc_change', and one that includes predicted probabilities for 'Model 5' as shown in Figure 5.

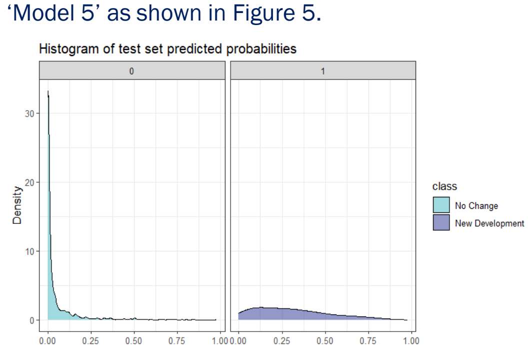

Figure 5. Histogram of test set predicted probabilities

ROC Curve is conducted to evaluate the performance of the model and the Area Under the Curve (AUC) is 0.9118

A cross-validation is also conducted to estimate and validate the performance metrics including accuracy, specifity, and sensitivity. Figure 6 represents the cross-validation model fit metrics

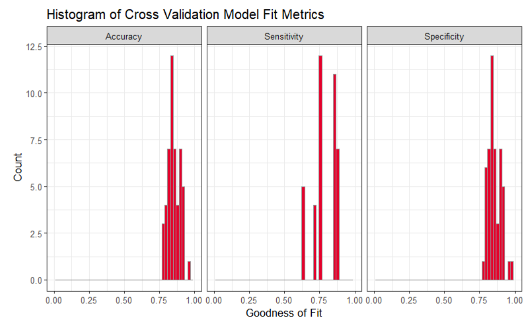

Figure 6. Cross-validation Model Fit Metrics

USE-CASE SCENARIOS

The model is then used to predict two use-case scenario that may occur in the year 2031.

First scenario is to predict the development area in 2031 and distributing 'new'population to those area. Figure 7 shows the prediction

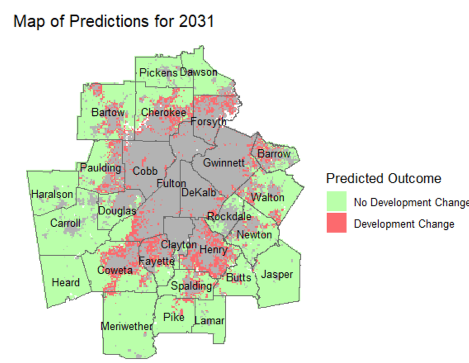

Figure 7. Prediction of Development Area in 2031

The second scenario is to predict the development area in 2031 if there will be a new highway in Atlanta MSA as shown in Figure 8.

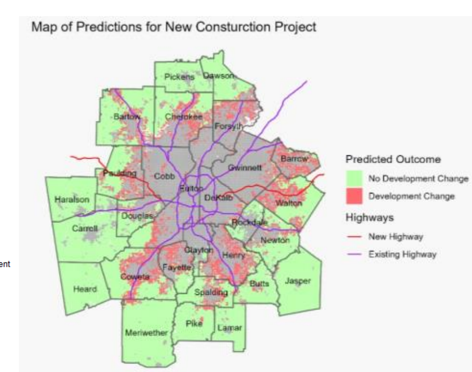

Figure 8. Prediction of Development Area in 2031 as the effect of New Highway Infrastructure